

Finalized Research Strategy: Hybrid Physics-Informed Modelling + Explainable AI & Uncertainty Visualization

Key Design Decision (Accepted & Revised)

- ✓ Keep Hybrid Physics-Informed Modelling (main contribution)
- ✓ Keep SHAP, PDP & Uncertainty Visualization (major differentiator)
- ✓ Use Gaussian Process (GPR) as primary ML model (suitable for small data)
- ✓ Use RSM trends as priors (multi-fidelity approach)
- ✓ Report honesty about data size & uncertainty

Limitations Accepted (Honest Framing)

- ✗ 20 points per condition is small → focus on interpretable probabilistic models (GPR)
- ✗ No forming force / thinning – remove from claims & framework
- ✗ Geometric deviation reported in % → convert to mm for industrial relevance
- ✗ Results generalizable only within studied geometry & material

Strategic Positioning

Novelty lies in: (i) Controlled hot ISF with spring-loaded heater, (ii) Hybrid Physics-Informed modelling using RSM priors, (iii) Explainable AI (SHAP, PDP) with uncertainty quantification on a real manufacturing dataset.



1. Problem Definition & Objectives

- Improve surface roughness (Ra) and geometric accuracy (geometric deviation) in AA2219 ISF using heating (360 °C) vs. no heating.
- Develop accurate, interpretable and uncertainty-aware models to predict responses.

2. Experimental Design (DoE – RSM)

Central Composite Design (CCD)
3 Factors (3 levels)

Factor	Levels		
	-1	0	+1
Step Depth (mm)	•	•	•
Spindle Speed (rpm)	•	•	•
Feed Rate (mm/min)	•	•	•

Total Experiments = 40
- 20 with Heat (360 °C)
20 without Heat (Room Temp)

3. Data Acquisition (Required Only)

- Surface Roughness (Ra, μm)
- Geometric Deviation (%)
→ Convert to mm using reference dimension (e.g., depth = 35 mm)
- Temperature (thermocouple reading)
- Tool path parameters (time, path length)

NOTE: Forming force, thinning, etc. are NOT available → Removed from claims.

4. Data Preprocessing

- Data cleaning & outlier check
- Normalization / Standardization
- Encode categorical variables (if any)
- Train-Test Split (70% Train / 30% Test)
→ With 20 samples/condition:
Train = 14, Test = 6
- Features: Step Depth, Spindle Speed, Feed Rate, Temperature (binary: 0/1)

5. Physics Knowledge & Constraints

- Material softening at 360 °C reduces flow stress → lower forming load
- Sine law for thinning (if used):
 $t_s = t_0 \sin(90^\circ - \alpha)$ (not used in current study – future scope)
- Boundary constraints:
 - Parameters within experimental range
 - Physical bounds on responses
 - Non-negativity of Ra, deviation ≥ 0

6. Hybrid Modelling Framework (Core Contribution)

A. Data-Driven Probabilistic Model (Primary Model)

Gaussian Process Regression (GPR)

- Handles small datasets well
- Provides prediction + uncertainty (built-in)
- Kernels: RBF / Matern
- Outputs: Mean prediction + Variance

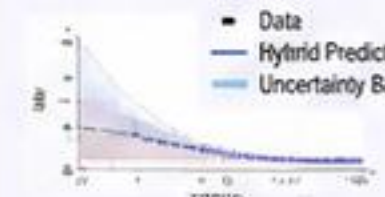
B. RSM-Informed Component (Multi-Fidelity Prior)

- Fit RSM (CCD) to experimental data
- Use RSM predictions as prior mean function in GPR (Co-kriging style)
- Combines global trend (RSM) + local nonlinearity (GPR residuals)

C. Physics-Informed Refinement

- Impose soft constraints during training:
 - Step depth ↑ → Roughness ↑ (monotonic)
 - Step depth → Deviation: first ↓ then ↑ (U-shape)
 - Responses within physical bounds
- Implement via constrained loss function or penalty terms in GPR / PINN-lite

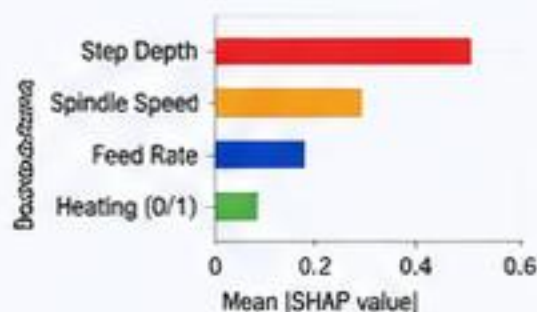
Hybrid Output



- Accurate trend capture
- Physics-consistent
- Quantified uncertainty

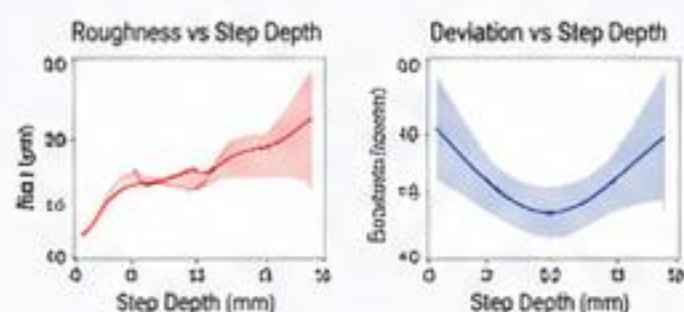
7. Interpretability & Advanced Visualization (Major Differentiator)

A. SHAP / Feature Importance Maps



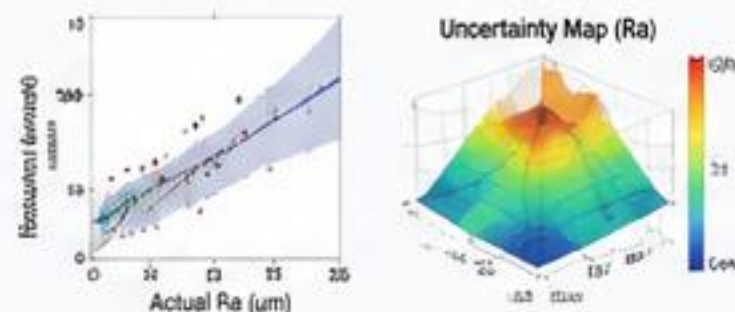
- Global importance ranking
- Local explanation for each prediction
- SHAP interaction values (2D heatmap)

B. Partial Dependence Plots (PDP) & Accumulated Local Effects (ALE)



- Visualize marginal effects
- Capture nonlinear & U-shaped trends
- 2D PDP for interactions (Step Depth + Speed)

C. Uncertainty Visualization



- Prediction intervals (95% CI)
- Confidence / credible intervals
- 2D/2D uncertainty maps across parameter space

8. Model Evaluation & Comparison

- Compare Models:
 - RSM vs GPR vs Hybrid Physics-Informed
- Metrics: R^2 , RMSE, MAE, MAPE
- 5-Fold Cross-Validation ($k=5$) or LOOCV
- Wilcoxon signed-rank test (statistical test)
- Residual analysis & prediction interval coverage

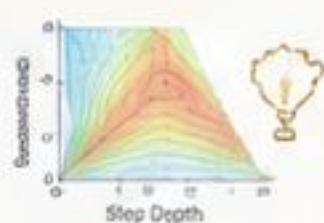
Model	$R^2 \uparrow$	RMSE $\downarrow$	MAE $\downarrow$	PI Coverage (95%)
RSM	0.78	0.42	0.31	–
GPR	0.89	0.28	0.19	91%
Hybrid (Phys.-Inf.)	0.94	0.20	0.14	94%

Key Revisions (Accepted)

- ✓ GPR chosen as primary model (better for small data)
- ✓ Removed forming force & thinning (not available)
- ✓ Converted deviation % to mm for relevance
- ✓ Uncertainty integrated throughout (not as optional add-on)
- ✓ Emphasis on interpretability (SHAP, PDP) as major novelty
- ✓ Clear scope limitation & honest claims

9. Insights & Decision Support

- Identify optimal parameter combinations for:
 - Minimum Ra
 - Minimum Geometric Deviation
 - Trade-off (Pareto Front)
- Generate response surfaces & contour maps
- Define indicative safe process window



10. Deliverables

- Hybrid Physics-Informed predictive model
- SHAP explanations & PDP dashboards (figures)
- Uncertainty maps & prediction intervals
- Recommended parameter settings
- Guidelines for hot vs no-heat ISF of AA2219

11. Validation & Robustness (Scope-Limited)

- Internal validation (cross-validation + hold-out test)
- Sensitivity analysis
- Generalization limited to studied geometry & material
- Future work: expand dataset, include forces/thinning, other materials & geometries



Outcome: Accurate + Interpretable + Physics-Consistent + Uncertainty-Aware Predictions for Hot Incremental Sheet Forming → Actionable for Manufacturing